

Covariance-Method Claims Are Ensemble-Relative: Named Measures and an Audit Program

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Abstract

A claim about the behavior of a covariance or correlation method is a claim about a distribution over matrices, and the distribution is rarely stated. We measure how much that omission matters. Across fifteen named, seeded ensembles at identical dimension and sample size, the Ledoit–Wolf shrinkage intensity ranges from 0.07 to 0.86, and eleven audited claims from eleven fields flip verdict as the ensemble changes, for stated structural reasons. The `randomcov` library names the measures and seeds the draws, so any claim can be re-run under all fifteen; the results assemble into a matrix of claims against measures, and a joint leaderboard of scikit-learn and `precise` estimators has five different winners. The pattern survives scale: six of seven claims re-run from $n = 30$ to $n = 3000$ keep their verdict, and the seventh, hierarchical risk parity, flips under its own home-turf ensemble as dimension grows. Eleven audits are completed here; a docket from astronomy, genomics, neuroscience, and chemometrics, including ensemble-invariant null controls, is assembled for the rest.

1 The unstated measure

Empirical claims about covariance methods have the form “estimator A beats estimator B ,” “the shrinkage intensity is typically moderate,” “the factorization recovers structure,” with performance averaged over random test matrices. That average is taken with respect to some generating law, and in practice the law is inherited from convenience and from the field. Finance validates on equity panels; econophysics on the same panels with random-matrix nulls; geostatistics on smooth kernels; psychometrics on low-rank factor models; graphical modeling on sparse precisions; cosmology on mock catalogs of ordered correlation-function bins.

Different laws produce matrices with different spectra, different locality, and different factor content. A verdict certified under one law therefore carries no warrant under another. What the measurements below show is that this omission is not a mere technicality.

2 Precedents

We are hardly the first to notice that method comparisons inherit their verdicts from the simulation design; this is established already in several literatures. Morris et al. [2019] require

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simulation studies to declare the data-generating mechanism as a design choice of equal rank with the estimand. Boulesteix et al. [2013] trace the over-optimism of self-reported comparisons to test beds chosen by the method’s own authors. Nießl et al. [2022] re-analyze one benchmark under hundreds of design and analysis options and find that almost any method can reach almost any rank. Hooker [1995] made the complaint for heuristics; Wolpert and Macready [1997] give the limit statement, that averaged over all objectives all optimizers tie.

The correlation-matrix version also has precedent. Joe [2006] and Lewandowski et al. [2009] supply the standard samplers. Hardin et al. [2013] argue that the habitual generators produce unrealistic matrices and bias the evaluation of statistical techniques, and propose a better one. Marti [2020] learns a generator from financial correlation matrices and reports, in accompanying experiments, that the HRP advantage fades on matrices without hierarchical structure, a result restated in Marti et al. [2021]. Papenbrock et al. [2021] evolve synthetic correlations to ask which correlation structures favor which allocator. Alvarez et al. [2014] document the Bayesian mirror image: an inverse-Wishart prior biases the posterior variance upward and the correlation toward zero when the true variance is small relative to the prior mean, and the bias persists as the sample grows.

What the present paper adds is the instrument and the audits: a fixed battery of fifteen named, seeded, public measures; the re-execution of eleven published claims from eleven fields under all of them; the claims-by-measures matrix as the reporting format; and provably invariant theorems run as null controls.

3 Fifteen named ensembles

`randomcov` generates correlation matrices under fifteen named, seeded measures, and the names describe the constructions: `lkj`, `onion`, `vine` (elliptope samplers; Joe, 2006, Lewandowski et al., 2009); `wishart`, `spectrum` (prescribed spectral laws); `factor`, `hierarchical`, `block_equicorr` (low-rank and clustered structure); `ar1`, `kernel` (banded and locality-driven fields); `sparse_precision` (graphical-model structure); `archakov_hansen` (log-matrix parameterization; Archakov and Hansen, 2021); `residuals`, `walk`, `animals` (empirical-flavored and trajectory ensembles).

Variance vectors are drawn separately, so correlation structure and scale can be crossed independently. Every generator accepts an explicit seed, and every number in this paper descends from one; there is no number here we cannot regenerate. The scripts sit beside the paper.

Three of the fifteen are constructions of one measure. `lkj`, `onion` and `vine` each sample $LKJ(\eta)$ by a different route, and they agree: across the eleven audits below their verdicts match to within sampling noise, and they rank seventeen estimators in near-identical order. That agreement is a check on the instrument. When three routes to one law return one answer, a disagreement between laws is a property of the laws.

4 Two demonstrations

We begin with the Ledoit–Wolf shrinkage estimator because it is used across so many applications and because its diagnostic is easy to read: the intensity says how far the sample covariance should be pulled toward a target. Fix $n = 30$ variables, each observed $T = 60$

ensemble	intensity	sd
residuals	0.072	(0.019)
animals	0.072	(0.013)
factor	0.101	(0.038)
archakov_hansen	0.120	(0.022)
kernel	0.132	(0.082)
walk	0.199	(0.038)
block_equicorr	0.247	(0.096)
vine	0.355	(0.022)
lkj	0.363	(0.024)
onion	0.363	(0.030)
hierarchical	0.369	(0.070)
ar1	0.469	(0.226)
wishart	0.506	(0.032)
spectrum	0.532	(0.058)
sparse_precision	0.862	(0.066)

Table 1: Ledoit–Wolf shrinkage intensity by ensemble at $n = 30$, $T = 60$: mean and standard deviation over twenty seeded draws.

times, and draw twenty panels per ensemble. This is not close to the $n = T$ boundary, and Table 1 shows the variation anyway.

The intensity spans 0.072 to 0.862 at identical (n, T) . A paper reporting “typical” shrinkage from one generator is reporting a property of its generator. The ordering is interpretable: strong low-rank structure (**animals**, **residuals**, **factor**) wants little shrinkage, and near-unstructured draws (**wishart**, **sparse_precision**) want a great deal. The within-ensemble standard deviation is informative too; **ar1** spans 0.23 across its own parameter draws.

The second demonstration is software of our own. A deterministic algorithm for multinomial probit choice probabilities (the **winning** package; paper in that repository) reports median absolute error by ensemble after fitting a structured covariance grammar. For twelve of the fifteen measures the error is $1\text{--}4 \times 10^{-5}$; for the dense, strongly correlated ensembles it is $1\text{--}4 \times 10^{-4}$, and under **kernel** fields 4×10^{-3} , because the fit does not yet admit the band-diagonal structure a kernel covariance calls for. A validation run on any single ensemble would have reported either “essentially exact” or “inadequate,” both correctly, for the same method. That experience was much of the motivation for writing the generator library.

5 Eleven audits, eleven fields

The same brittleness might apply more broadly, and where it does there is room to improve existing tools. The method is short. We take a published claim, re-run its demonstration under every generator in the library, and report the verdict.

We audit one claim from each of eleven fields: seven about estimators and constructions (Table 2) and four about inference and dimensionality (Table 3). Figure 1 assembles the results into a single matrix and outlines each claim’s *home turf*, the correlation structure it was demonstrated on or intended for, as best we can tell. All runs use $n = 30$ with twenty replications unless stated. Sampled panels use $T = 60$ except where the claim’s own regime dictates otherwise.

Finance. We start here because it is the field we know best. López de Prado [2016] reports that hierarchical risk parity delivers lower out-of-sample variance than a long-only, fully-invested quadratic minimum-variance optimizer, and names estimation error in the inverse as the culprit. We build both portfolios from the same sample covariance and score the true variance $w^\top \Sigma w$ under the generating covariance. Column 1 of Table 2 gives the median ratio of HRP to long-only minimum variance.

HRP wins on every draw under **sparse_precision** and splits draws roughly evenly under **ar1** and **hierarchical**, the ensembles whose dependency structure the bisection tree can match; it loses by an order of magnitude or more under **animals**, **factor**, and **archakov_hansen**, where the true matrix rewards positions a long-only tree cannot take. Against unconstrained minimum variance the losses grow by many more orders of magnitude: the median ratio is $\sim 10^{8.7}$ under **walk** and $\sim 10^{11.2}$ under **residuals**, because near-singular ensembles admit near-perfect hedges that only signed weights can reach. Both of those ensembles reach the positive-semidefinite floor, and the exponent scales inversely with the clip, falling a decade for every decade the clip loosens. The ensemble sets the direction of the result and the floor sets its size.

Econophysics. Laloux et al. [1999] find the bulk of the sample correlation spectrum indistinguishable from Marchenko–Pastur noise. The cleaning recipe that follows, flattening every eigenvalue below the edge $(1 + \sqrt{n/T})^2$ and reporting the result, is Laloux et al. [2000]. Column 2 gives the Frobenius-loss ratio of clipped to raw sample correlation. The cleaning is excellent under **sparse_precision** (0.30) and **hierarchical** (0.76), and it damages the estimate under eleven of fifteen ensembles, worst at 1.43–1.77 under **animals**, **kernel**, **archakov_hansen**, and **residuals**.

The mechanism is not quite the textbook one. **hierarchical** does carry a spike over a bulk. **sparse_precision** carries none, with a largest true eigenvalue of 1.43 against an edge of 2.91, and clipping wins there because the sub-edge spectrum is nearly flat already, so flattening it costs almost nothing. Under the four worst ensembles the sub-edge mass is dispersed, and flattening erases genuine structure as noise. What predicts the verdict is the dispersion of the true spectrum below the edge. The recipe presumes that signal means spikes, and locality puts signal in the bulk.

Geostatistics. The screening effect of Stein [2002] says that a few nearest neighbors, here the most correlated, nearly attain the full kriging variance. This is a population property, so no sampling is involved. Column 3 reports the median ratio of conditional variance given the best five predictors to conditional variance given all $n - 1$. Screening holds to within a few per cent under **ar1** (Markov, ratio 1.00), **hierarchical**, and **block_equicorr**. It also reads 1.00 under **sparse_precision**, but vacuously: conditioning on all $n - 1$ neighbors removes only about four per cent of the variance there, so there is nothing for five neighbors to miss.

Screening fails several-fold under the elliptope samplers **lkj**, **onion** and **vine**, and under **kernel** itself (5.7). Every **kernel** draw is squared-exponential, whose spectral density decays faster than any algebraic rate and so falls outside Stein’s conditions; interpolation is nearly deterministic from all sites but not from five. The failure tracks the drawn length scale almost perfectly (correlation 0.98 between length scale and log ratio), and a Matérn-3/2 field on the identical point clouds gives 1.04 instead. That is Stein’s own caveat, that screening depends on smoothness, reproduced as a contrast.

Psychometrics. Parallel analysis [Horn, 1965] retains the components whose sample eigenvalues exceed those of independent data of the same shape. Horn compared against the mean null eigenvalue; we use the now-standard 95th-percentile form of Glorfeld [1995]. It is the standard answer to “how many factors?” Column 4 gives the mean retained count: 1.8 under **sparse_precision** to 7.3 under **onion** at identical (n, T) , more than a fourfold spread for what is supposed to be an objective count.

The same sweep scores the eigenvalue-greater-than-one rule of Kaiser [1960] against the population count of eigenvalues above one. Kaiser over-extracts under **factor** (5.7 retained against 3.0 true), under-extracts under **wishart** (11.2 against 13.1), and disagrees with parallel analysis by a factor that is itself ensemble-dependent: 1.4 under **residuals**, 7 under **sparse_precision**. The number of factors a study reports is therefore, in part, a property of the generating measure its data happen to resemble.

Data assimilation. Ensemble Kalman filtering runs with far fewer members than state variables, and the standard remedy is covariance localization: taper the sample covariance entrywise with a Gaspari–Cohn function [Gaspari and Cohn, 1999] of inter-variable distance [Houtekamer and Mitchell, 2001, Hamill et al., 2001]. Paz and Sánchez [2015] apply the closely related covariance tapering of Kaufman et al. [2008] to correlation-function bins, with a Wendland taper and a two-taper precision estimator. The two share the assumption this audit tests, that the taper coordinate carries the truth’s locality, so the audit applies to both fields. With $N = 20$ members for $n = 30$ variables and a taper of index distance with half-radius 6, column 5 reports the Frobenius-loss ratio of tapered to raw.

The split is interpretable. Localization helps wherever the estimate is noisy and the truth carries no long-range structure in taper coordinates, including under the dense ellipsope samplers, where variance reduction beats the bias. It hurts under **factor**, **animals**, and **residuals** (1.66–1.84), whose common factors are long-range by construction. It also hurts under **kernel** (1.46), the one ensemble that is local by construction, because kernel locality lives in a latent two-dimensional point cloud rather than in index order. Localization requires locality in the coordinates the taper can see.

Signal processing. Diagonal loading is the standard repair for the sample-covariance MVDR beamformer: Cox et al. [1987] derive it from a constraint on white-noise array gain, against sensor and steering perturbations, and Carlson [1988] tie it to covariance estimation error in the snapshot-limited regime. With $T = 40$ snapshots, a random unit steering vector, the distortionless constraint, and loading $\delta = 0.1 \text{tr}(\hat{\Sigma})/n$, column 6 reports the median ratio of true output power, loaded over unloaded. The picture is mixed. Loading roughly halves the output power under six of the fifteen ensembles, which is the win it is sold as, and multiplies it under **archakov_hansen** (20.5), **animals** (33), **kernel** (136), **residuals** ($10^{7.7}$), and **walk** ($10^{7.9}$). The mechanism is the hedge from the finance audit seen from the other side: under near-singular ensembles the beamformer’s null-steering lives in the smallest eigenvalues, which is exactly what loading erases.

Graphical modeling. The graphical lasso of Friedman et al. [2008] made the sparsity prior on the precision matrix cheap enough to be routine. Their own claim was speed; the field attached to it the promise that the prior improves estimation. That promise underwrites partial-correlation connectomes [Smith et al., 2011] and sparse cosmological precision matrices

ensemble	HRP/MV (fin.)	clip/raw (econo.)	near/full (geost.)	PA count (psych.)	taper/raw (assim.)	load/raw (sig. proc.)	gl/LW (graph.)
animals	8.38	1.43	5.53	3.3	1.66	32.8	1.00
ar1	0.98	1.20	1.00	4.5	0.53	0.48	0.69
archakov_hansen	71.8	1.47	131	4.7	1.41	20.5	1.00
block_equicorr	1.09	1.00	1.04	3.0	1.00	0.55	0.78
factor	12.3	1.38	1.06	2.2	1.69	0.94	1.09
hierarchical	1.00	0.76	1.02	2.3	0.82	0.45	0.68
kernel	1.51	1.46	5.69	4.1	1.46	136	1.00
lkj	2.67	1.29	10.6	7.3	0.79	2.26	0.99
onion	2.83	1.29	9.04	7.3	0.78	3.29	1.00
residuals	2.75	1.77	$\sim 10^{8.3}$	4.0	1.84	$\sim 10^{7.7}$	1.00
sparse_precision	0.76	0.30	1.00	1.8	0.50	0.48	0.63
spectrum	1.11	0.94	1.32	5.8	0.66	0.51	0.99
vine	2.68	1.29	11.9	7.0	0.81	4.13	1.00
walk	1.23	1.11	$\sim 10^{8.7}$	3.7	1.14	$\sim 10^{7.9}$	1.00
wishart	1.19	1.04	1.50	6.5	0.66	0.51	0.95

Table 2: Seven audited claims about estimators and constructions; all entries are medians over twenty seeded draws. Column 1: out-of-sample true-variance ratio, HRP over long-only minimum variance (below one: HRP wins). Column 2: Frobenius-loss ratio, Marchenko–Pastur-clipped over raw sample correlation (below one: cleaning helps). Column 3: population conditional-variance ratio, five best predictors over all (near one: screening holds). Column 4: mean components retained by parallel analysis. Column 5: Frobenius-loss ratio, Gaspari–Cohn-tapered over raw sample covariance at $N = 20$ members (below one: localization helps). Column 6: true beamformer output power, diagonally loaded over unloaded at $T = 40$ snapshots (below one: loading helps). Column 7: precision-matrix Frobenius-loss ratio, cross-validated graphical lasso over inverted Ledoit–Wolf (below one: the sparsity prior wins). Each column’s verdict flips across rows. **walk** and **residuals** are rank-deficient by construction and reach the positive-semidefinite floor, so their entries in the two inverse-sensitive columns (3 and 6) are set by that floor: the exponents scale inversely with the clip and would move decade for decade under a looser one. They establish failure and nothing more precise than that.

[Padmanabhan et al., 2016]. Column 7 reports the Frobenius loss of the cross-validated glasso precision against the true precision, relative to inverting a Ledoit–Wolf shrinkage estimate of the kind Schäfer and Strimmer [2005] recommend for gene networks.

The sparsity prior wins where precisions are sparse or banded, at 0.69 under **ar1**, 0.68 under **hierarchical**, and 0.63 under **sparse_precision**. It ties across the dense elliptope middle. It loses under **factor** (1.09), and under **residuals** it ties at 1.00 while winning on none of the twenty draws. Dense low-rank structure is the worst case for sparsity.

We also performed a smaller companion study. With oracle shrinkage intensities, the constant-correlation target of Ledoit and Wolf [2004b] beats the scaled-identity target of Ledoit and Wolf [2004a] on every draw under **sparse_precision** (median loss ratio 0.62) and wins on only three of twenty draws under **factor**, where the two targets are otherwise indistinguishable at a median ratio of 1.00. Even the choice between two targets from the same authors is ensemble-relative.

The remaining four claims concern hypothesis tests, rates and counts (Table 3).

Evolutionary biology. Cheverud [1996] declares two genetic covariance matrices similar when their mean response-vector correlation under common random selection gradients exceeds the null for independent random vectors. We apply the test to independent pairs drawn from the same ensemble. This is a population property with no sampling noise, so a sound test should declare similarity about 5% of the time. It is at or below nominal under **factor** and **archakov_hansen** (0%) and **animals** (10%), and declares independent matrices similar 100% of the time under eleven of the fifteen ensembles. Cheverud and Marroig [2007] concede as much in passing: in practice the null of no shared structure “is nearly always rejected.”

Under **sparse_precision** the mean skews correlation between independent matrices is 0.97, because near-diagonal matrices respond to any skewer with the skewer itself. Rohlf [2017] raises this objection in general terms.

Climatology. North et al. [1982] give a rule of thumb: when an eigenvalue’s sampling error $\lambda\sqrt{2/T}$ is comparable to the spacing to a neighbor, the sampling error in the associated EOF is comparable to the neighboring EOF. They draw the converse as a first-order corollary, that a stand-out eigenvalue has small EOF sampling error, and say plainly that the rule offers no guidance on where to truncate. It is the corollary we score.

We score the false-reassurance rate: among draws the rule declares separated, how often the leading sample eigenvector is mostly wrong (squared alignment with the truth below one half). The rate is 0% under six ensembles, 57% under **wishart**, and 100% under **sparse_precision**, which declares separation on only 12% of draws but is always wrong when it does. The rule is a first-order perturbation statement: excellent for spiked spectra, and unreliable exactly where spectra are quasi-degenerate.

Ecology. Guillot and Rousset [2013] report that the Mantel test [Mantel, 1967] rejects independence between distance matrices at 25–55% under strong spatial autocorrelation, at nominal 5%. Our audit puts the original claim and the critique on the same footing (two independent site variables drawn with covariance C with Euclidean distance matrices and 499 label permutations). The realized type-I error is the nominal 5% under **onion** and **vine** (and 6% under **lkj**), and 38–43% under **factor**, **kernel**, **animals**, and **residuals**. This reproduces the published inflation under locality-and-factor structure and shows calibration under dense ellipsope measures. “The Mantel test is broken” and “the Mantel test is fine” are both correct reports, from different ensembles.

Quantitative genetics. Kirkpatrick [2009] reports effective dimension counts $n_D = \sum_i \lambda_i / \lambda_{\max}$ of roughly one to two for published genetic covariance matrices, read as evolutionary constraint. At the low precision typical of breeding designs, $T = 15$ here, the estimated n_D deflates the truth by an ensemble-dependent factor. A true n_D of 12.3 under **sparse_precision** is reported as 5.2 (ratio 0.42), while **factor** and **animals** are nearly undistorted (0.90, 0.89) because their truth is already low-dimensional. Across all fifteen ensembles the estimate never exceeds 5.2, whatever the truth. A low reported n_D cannot by itself distinguish low-dimensional biology from a noisy estimate of diffuse structure.

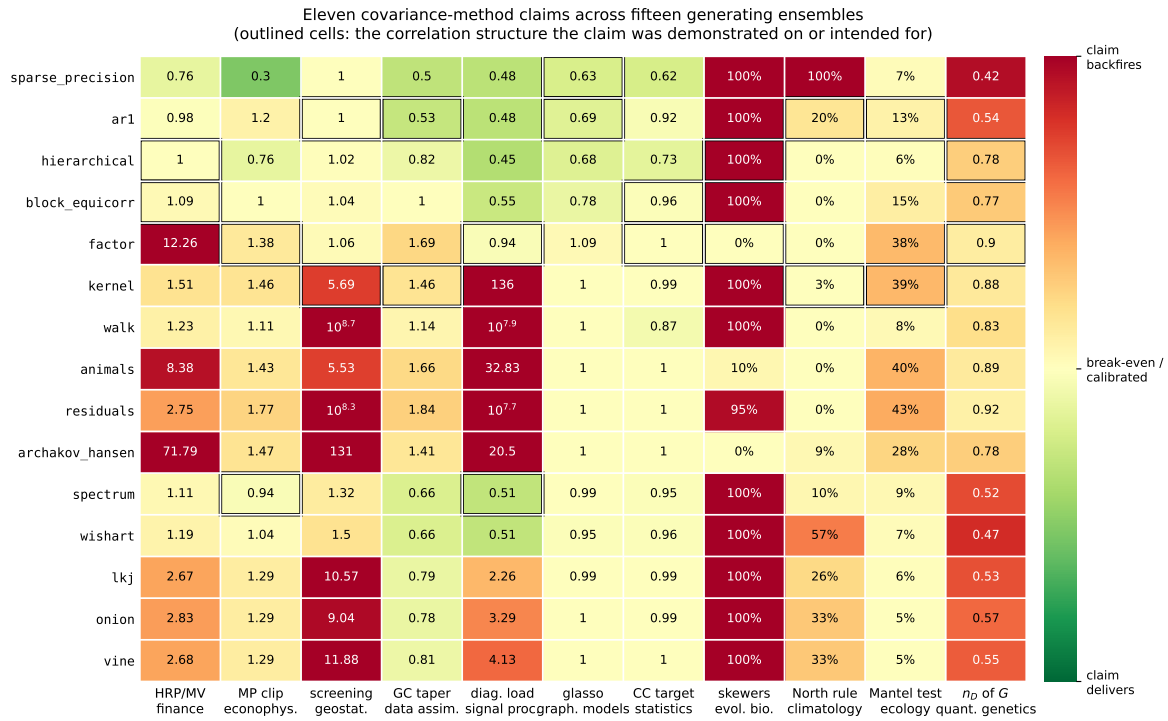


Figure 1: The audit matrix: eleven claims (columns, with their fields) against fifteen generating ensembles (rows, ordered roughly from structured to dense). Green: the claim delivers under that measure. Yellow: break-even or calibrated. Red: the claim backfires or the test is broken. Cell entries are the medians and rates of Tables 2 and 3. Outlined cells mark each claim’s home turf, the correlation structure it was demonstrated on or intended for.

ensemble	skewers false sim.	North rule false reass.	Mantel type-I	n_D ratio est/true
animals	10%	0%	40%	0.89
ar1	100%	20%	13%	0.54
archakov_hansen	0%	9%	28%	0.78
block_equicorr	100%	0%	15%	0.77
factor	0%	0%	38%	0.90
hierarchical	100%	0%	6%	0.78
kernel	100%	3%	39%	0.88
lkj	100%	26%	6%	0.53
onion	100%	33%	5%	0.57
residuals	95%	0%	43%	0.92
sparse_precision	100%	100%	7%	0.42
spectrum	100%	10%	9%	0.52
vine	100%	33%	5%	0.55
walk	100%	0%	8%	0.83
wishart	100%	57%	7%	0.47

Table 3: Four audited claims about inference. Column 1: share of independent matrix pairs that random skewers declares similar (sound: $\approx 5\%$). Column 2: among draws North’s rule declares separated, share whose leading EOF is mostly wrong (squared alignment below $\frac{1}{2}$). Column 3: realized type-I error of the Mantel test at nominal 5%. Column 4: estimated over true effective dimensions of a covariance matrix at $T = 15$ (undistorted: near one).

6 Do claims scale?

Every study above fixes $n = 30$. A reasonable objection is that ensemble-relativity is a small-panel artifact, that estimation noise dominates at $n = 30$ and would wash out with more assets. We re-run the Ledoit–Wolf intensity demonstration of Section 4 and five of the seven Table 2 claims (finance, econophysics, data assimilation, signal processing, and the graphical lasso, both the cross-validated penalty and a fixed one) across a 5×5 grid, $n \in \{30, 100, 300, 1000, 3000\}$ crossed with $T/n \in \{0.5, 1, 2, 4, 8\}$, medians over twenty-four seeded draws per cell (twelve at $n = 1000$ and $n = 3000$, for cost). Figure 2 plots all thirty-five cells of every audit, one line per ensemble. The two largest passes cost on the order of a hundred worker-hours apiece.

The spread does not close. At $n = 3000$, $T/n = 1$, the Ledoit–Wolf intensity still ranges from 0.002 (**factor**) to 0.780 (**sparse_precision**), a wider ratio than the 0.157–0.720 spread we saw at $n = 30$. Marchenko–Pastur clipping helps **sparse_precision** more as n grows ($0.293 \rightarrow 0.148$ at $T/n = 2$) and hurts **factor** far more ($1.366 \rightarrow 9.432$); its verdict does not flip for a single literature ensemble between $n = 30$ and $n = 3000$ at fixed $T/n = 2$, and neither does diagonal loading’s, the graphical lasso’s, or Gaspari–Cohn tapering’s.¹ So for six of the seven claims re-run, a claim’s ensemble-verdict at $n = 30$ is a reliable predictor of its verdict at a hundred times the dimension.

The exception is hierarchical risk parity, but only under one of the two ensembles that looked exceptional at $n = 30$. At $n = 30$, $T/n = 2$, HRP roughly breaks even against long-

¹The fixed-penalty graphical lasso records one nominal flip, **lkj** from 1.004 to 1.000, indistinguishable from a tie. The cross-validated graphical lasso is incomplete at $n = 1000$, roughly half its cells missing to per-cell timeouts, and dropped entirely at $n = 3000$, where a single fit runs into the hour.

only minimum variance under **ar1** and **hierarchical**, the two literature ensembles whose structure the bisection tree is built to match (ratios 0.890, 0.962, twenty-four draws). At $n = 3000$, the twelve-draw grid median moves to 1.259 and 2.580, both apparent losses. A targeted replication at forty-eight draws confirms the **hierarchical** loss (median 2.557; every single draw favors minimum variance) and retracts the **ar1** one: the forty-eight-draw median is 0.914, 56% of draws still favor HRP, and the ratio ranges from 0.69 to 1.89.² The twelve-draw estimate had landed on the high side of a noisy statistic. A rep count that is fine for most cells is not automatically fine for all of them.

sparse_precision remains an outright HRP win at every scale we tested, and **ar1** remains close to a tie at every scale we tested. Only **hierarchical** flips, and the cause is the ensemble rather than the algorithm: its cluster correlation is a function of absolute cluster size, so as n grows the matrix thins toward plain equicorrelation: its non-equicorrelation content falls fourfold from $n = 30$ to $n = 3000$, and the share of pairs correlated above 0.2 falls a hundredfold. The structure HRP is built to exploit is gone by $n = 3000$, while the long-only quadratic program keeps exact use of the full covariance regardless of n . A home-turf verdict is not guaranteed to survive scale, and a low rep count is not guaranteed to survive a check.

We checked the other close calls. At $n = 3000$, $T/n = 2$, three literature ensembles sit within 2% of the clip/raw = 1 threshold on the twelve-draw grid: **spectrum** (0.996), **kernel** (1.017), **archakov_hansen** (1.020), closer calls than HRP’s **ar1** ever was. A forty-eight-draw check confirms all three (medians 0.996, 1.018, 1.021; no range wider than $[0.98, 1.06]$).³ All three hold. The difference from HRP is the statistic: a Frobenius-norm ratio has almost no per-draw variance, while HRP’s long-only optimizer does, so twelve draws are trustworthy for one and not for the other. The fragility is a property of the HRP audit’s statistic.

Five of the fifteen ensembles turn numerically rank-deficient at the largest dimensions tested and need a positive-semidefinite floor before any audit runs: three package-native curiosities (**walk** on 58% of $n = 3000$ panels, **residuals** on 33%, **animals** on 17%) and two literature ensembles (**archakov_hansen** on 14%, **kernel** on 3%). The floor is applied before scoring, so the numbers above already reflect it; a reader citing an **archakov_hansen** or **kernel** verdict at $n = 3000$ should know that a seventh to a thirtieth of its draws needed rescue just to stay a valid correlation matrix.

7 One leaderboard per measure

The audits test one claim at a time; the same harness can just as easily rank whole libraries. We fit the six covariance estimators of scikit-learn and the eleven batch-applicable estimators of **precise** to the same twenty seeded panels per ensemble and rank them by median relative Frobenius loss against the generating covariance (Figure 3). Five estimators take first place somewhere: OAS on four ensembles, fixed shrinkage and the graphical lasso on three each, Ledoit–Wolf on two, and, perhaps surprisingly, the raw sample covariance on **animals**, **kernel** and **residuals**, three of the ensembles the intensity table already marked as wanting almost no shrinkage. The graphical lasso is first on the three precision-structured rows (**sparse_precision**, **ar1**, **hierarchical**) and twelfth of seventeen under **animals**, **residuals**, and **factor**, a large fall. The **precise** estimators are built for time-varying covariance and

²`hrp_n3000_replication.py`, beside this paper. Reps 0–11 reuse the grid’s own deterministic seeding; reps 12–47 extend it under the identical scheme.

³`clip_n3000_replication.py`, beside this paper, same seeding scheme as the HRP check.

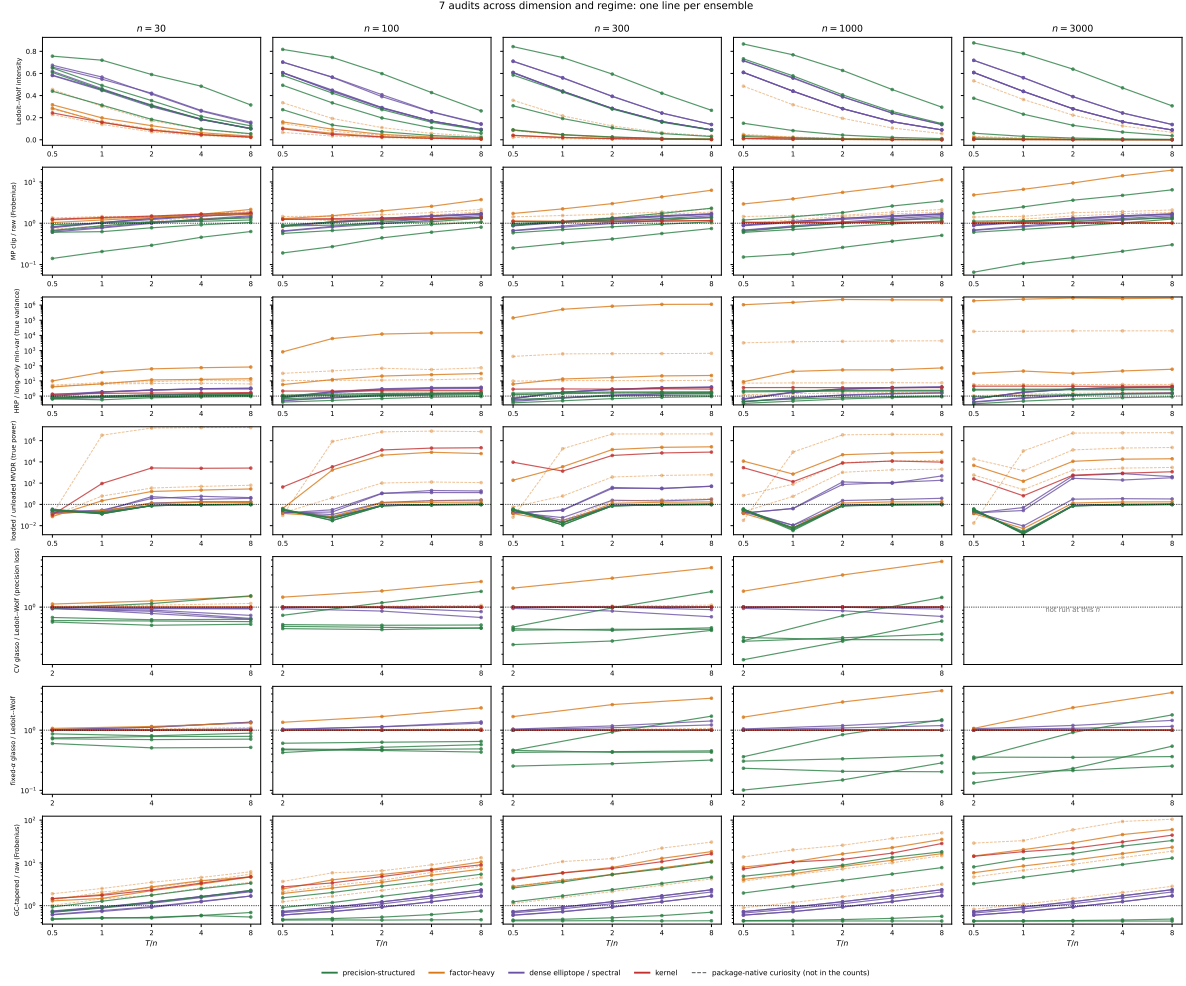


Figure 2: Seven audits (rows: Ledoit–Wolf intensity, MP clipping, HRP, MVDR loading, cross-validated and fixed-penalty graphical lasso, Gaspari–Cohn tapering) across five dimensions (columns, $n = 30$ to 3000) and five regimes within each panel ($T/n = 0.5$ to 8), one line per ensemble, colored by structural family; dashed lines are package-native curiosities. Rows share a y -axis scale within each audit so drift with n is visible. Cross-validated glasso is not run at $n = 3000$.

pay for that design on static panels, filling the lower ranks. A library default is a bet on a measure, whether or not the author meant it that way.

8 Scope

The audits measure the ensemble-sensitivity of a claim, often outside the scope its authors gave it. Stein proved that screening depends on smoothness; Bickel and Levina [2008] state the bandable class their banding rates require; López de Prado demonstrated HRP on equity covariances and said so; the data-assimilation literature has always tied localization to physical distance.

The selection of claims is ours, and it leaned toward methods we knew to be useful or hoped would be. A red cell in Figure 1 therefore says where a claim holds, and says nothing against the claim where its authors placed it. Several audits run a method well outside the regime it was built for, deliberately, because that is where second-hand use puts it.

The battery is a choice too. Fifteen named measures are not a uniform measure over correlation matrices, since no such thing exists, and they are not exhaustive. Adding a sixteenth ensemble can only add flips. Every verdict here is a statement about these fifteen laws, and the naming is what lets a reader add the sixteenth.

The risk arises downstream, in less careful renderings of the original verdict. A claim certified under one measure is quoted, cited, and defaulted into software as if it were measure-free, and the second-hand version of a scoped claim is usually unscoped. Naming the ensembles and shipping the tests is the remedy.

9 The pattern in the matrix

The pattern across Figure 1 is not random: each field’s signature claim performs best on the ensembles that resemble the data the field grew up on, with instructive exceptions where a home-turf cell fails, as when screening fails inside the kernel ensemble. `sparse_precision` is kind to hierarchical bisection, eigenvalue cleaning, localization, and the graphical lasso at once, yet breaks random skewers and North’s rule. The dense elliptope samplers (`lkj`, `onion`, `vine`) are kind to almost nothing except Gaspari–Cohn localization, which delivers on all three (0.78–0.81), and the Mantel test, which they hold at or near its nominal level; they are also the most common default test bed. The same mechanism recurs under different names: the hedge that unconstrained minimum variance exploits is the null-steering that diagonal loading erases.

10 The docket

A survey of prominent covariance-method claims across further fields yields a docket for the next round, kept beside this paper. Sorting it produces a taxonomy. Claims that impose structure on the truth are ensemble-relative; theorems about Wishart sampling noise hold for every true covariance and cannot flip.

The structural claims are the audit targets. Paz and Sánchez [2015] taper clustering-statistic covariances, declining to assume the far off-diagonal entries are negligible, though the taper still presumes the bin coordinate carries the locality; Padmanabhan et al. [2016] estimate

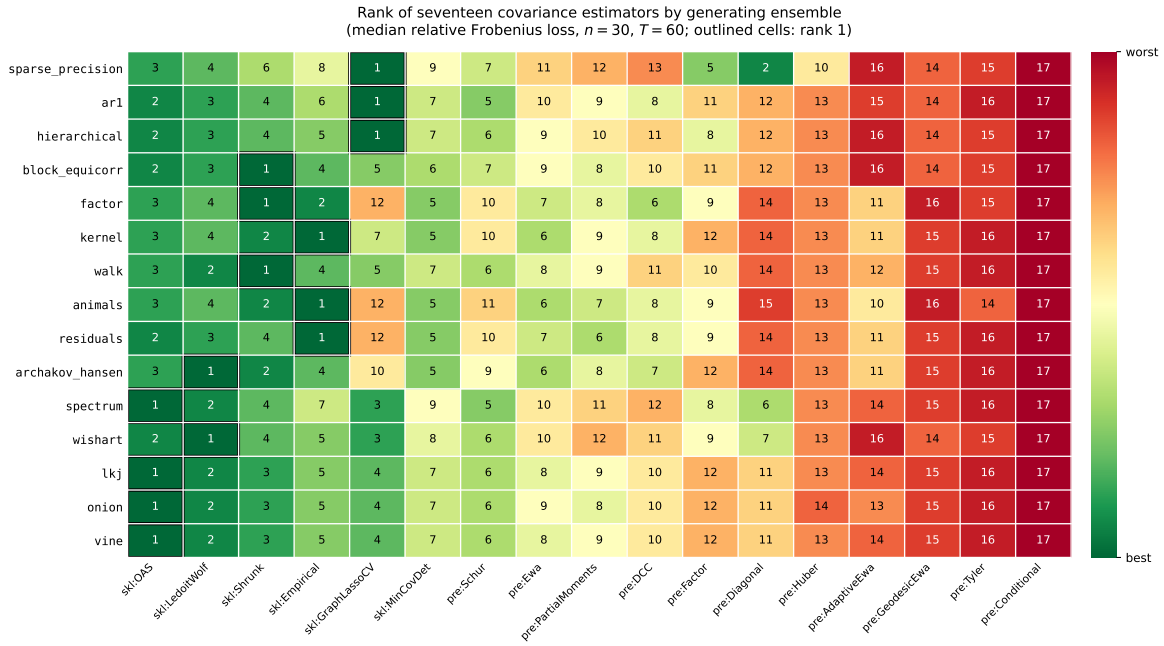


Figure 3: The leaderboard matrix: seventeen covariance estimators, six from scikit-learn and eleven from **precise**, ranked within each generating ensemble by median relative Frobenius loss at $n = 30$, $T = 60$, twenty seeded draws per ensemble. Rank-one cells are outlined; columns are ordered by mean rank.

sparse cosmological precisions and presume a sparse graph. Zhang and Horvath [2005] select gene-network powers by a scale-free-topology criterion that is itself, whether or not the authors framed it this way, a statement about the generating measure of gene correlation matrices. Smith et al. [2011] demonstrated the win of partial over full correlation for network detection on sparse simulated networks, and report the ordering reversing at fifty nodes. Norberg et al. [2009] judged jackknife covariances for galaxy surveys, and the verdict there hinges on cross-region correlation structure.

A second strand turns on spectral shape instead. Rule N [Preisendorfer, 1988] truncates EOFs by comparing sample eigenvalues against the null spectrum of independent data of the same shape, that is, against an ensemble. Ledoit and Wolf [2012] show the edge of nonlinear over linear shrinkage growing with spectral dispersion; Fan et al. [2013] build POET on pervasive factors plus sparse residuals; Frank and Friedman [1993] show the PCR-and-PLS-versus-ridge contest turning on whether the true coefficient vector aligns with the high-spread directions of the predictor eigenstructure. Each carries a predicted flip.

The invariant theorems serve as null controls. Hartlap et al. [2007] debias the inverse sample covariance exactly, for every true covariance; Dodelson and Schneider [2013] derive a parameter-variance inflation that, in the Gaussian limit, depends only on counts; Sellentin and Heavens [2016] marginalize the estimated covariance exactly under Wishart sampling; Lowry et al. [1992] prove the MEWMA chart depends on a known covariance only through a noncentrality parameter.

An audit pipeline that finds ensemble-dependence in the Hartlap factor has a bug, so we ran that control: with one fixed covariance per ensemble and four hundred panels at $(n, T) = (30, 60)$, the trace of the raw average inverse sample covariance overshoots the true inverse trace by a factor of 2.07 to 2.13 under every ensemble, against the Wishart factor $(T - 1)/(T - n - 2) = 2.11$, and the corrected ratio is 0.98 to 1.01 everywhere. The pipeline reports no ensemble-dependence where none can exist. The most instructive failures are invariant theorems wrapped in ensemble-relative plug-in practice, as when those corrections are applied to shrunk or tapered estimates whose distribution is no longer Wishart.

11 Closing

There is no free lunch. No method in the tables dominates across the rows. Each owns the ensembles whose structure it assumes, and the matrix draws the borders. We leave to future work the remaining docket entries, beginning with the nonlinear-against-linear shrinkage comparison and the WGCNA scale-free criterion.

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